

Decision Tree Example

Dataset 1

ID	Income level	Owns property	Has co-signer	Credit history	Default
1	low	TRUE	FALSE	fair	No
2	high	TRUE	TRUE	good	No
3	mid	FALSE	TRUE	poor	Yes
4	high	FALSE	FALSE	good	No
5	low	FALSE	FALSE	poor	Yes
6	mid	TRUE	TRUE	good	No
7	low	TRUE	FALSE	poor	Yes
8	high	TRUE	TRUE	fair	No
9	mid	FALSE	FALSE	fair	Yes
10	low	FALSE	TRUE	good	No

Tasks: Apply the **Decision Tree Learning Algorithm** to the dataset.

1. Calculate the **Entropy of the target attribute**.
2. Compute the **Information Gain** for each attribute
3. Draw the final Decision Tree.
4. Extract all decision rules from the constructed tree.
5. Predict whether a customer will default if:
 - Income Level = Mid
 - Owns Property = FALSE
 - Has Co-signer = TRUE
 - Credit History = Poor

Dataset 2

ID	Outlook	Temperature	Humidity	Wind	Play
1	sunny	hot	high	weak	No
2	sunny	hot	high	strong	No
3	overcast	hot	high	weak	Yes
4	rainy	mild	high	weak	Yes
5	rainy	cool	normal	weak	Yes

6	rainy	cool	normal	strong	No
7	overcast	cool	normal	strong	Yes
8	sunny	mild	high	weak	No
9	sunny	cool	normal	weak	Yes
10	rainy	mild	normal	weak	Yes
11	sunny	mild	normal	strong	Yes
12	overcast	mild	high	strong	Yes

Tasks: Apply the **Decision Tree Learning Algorithm** to the dataset.

1. Calculate the **Entropy of the target attribute**.
2. Compute the **Information Gain** for each attribute
3. Draw the final Decision Tree.
4. Extract all decision rules from the constructed tree.
5. Predict whether **Play = Yes or No** for the following instance:
 - Outlook = Sunny
 - Temperature = Mild
 - Humidity = Normal
 - Wind = Strong

Dataset 3

ID	Age	Years at company	Department	Remote work	Salary (k)	Attrition
1	29	1.5	Sales	TRUE	42	Yes
2	45	12	Engineering	FALSE	95	No
3	31	2	Marketing	TRUE	50	Yes
4	38	6	Engineering	FALSE	80	No
5	26	0.8	Sales	TRUE	38	Yes
6	50	20	HR	FALSE	70	No
7	34	3	Marketing	FALSE	55	No
8	41	9	Engineering	TRUE	88	No
9	27	1	HR	TRUE	40	Yes
10	44	11	Sales	FALSE	75	No

Tasks: Apply the **Decision Tree Learning Algorithm** to the dataset.

1. Calculate the **Entropy of the target attribute**.
2. Compute the **Information Gain** for each attribute
3. Draw the final Decision Tree.
4. Extract all decision rules from the constructed tree.
5. Predict whether an employee is likely to leave the company (**Attrition = Yes/No**) for the following instance:
 - Age = 30
 - Years at Company = 2
 - Department = Sales
 - Remote Work = TRUE
 - Salary = 45k